



Microprocessors And Microcontrollers

Assignment- Week 0

TYPE OF QUESTION: MCQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

The smaller unit of binary data

- a) Nibble
- b) Bit
- c) Byte
- d) word

Correct Answer: b

Detailed Solution: Bit: The smallest unit of binary data is a bit (short for "binary digit"). A bit can have only two possible values: 0 or 1. It is the fundamental building block of binary data in computing.

Nibble: A nibble is a group of 4 bits. For example, 1010 is a nibble.

Byte: A byte consists of 8 bits. It is the standard unit of data storage in most computer systems.

Word: A word refers to a data unit consisting of a fixed number of bits, depending on the computer architecture. For example, in a 32-bit system, a word is 32 bits, and in a 64-bit system, it is 64 bits.

QUESTION 2:

How many possible values for digital signals

- a) 4
- b) 2
- c) 8
- d) 1



Correct Answer: b

Detailed Solution:

Digital signals are based on the binary number system, which has only **two possible states**:

0 and **1**.

These two states represent the two voltage levels typically used in digital electronics:

0 (Low or Off)

1 (High or On)

This binary nature of digital signals allows for reliable representation and transmission of data in computers and digital devices

QUESTION 3:

Which number system has a base of 16

- a) Decimal
- b) Octal
- c) Hexadecimal
- d) Binary

Correct Answer: c

Detailed Solution:

Number systems are categorized based on their **base** (or radix), which indicates the number of unique digits, including zero, used in the system.

Hexadecimal (Base 16):

The **hexadecimal number system** has a base of **16**, meaning it uses 16 unique symbols: **0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F**.

Here, the letters **A** to **F** represent the decimal values **10 to 15**, respectively.

Decimal (Base 10):

The decimal system is the standard number system with a base of **10**. It uses the digits **0–9**.

Octal (Base 8):

The octal system has a base of **8**, using the digits **0–7**.



Binary (Base 2):

The binary system has a base of **2**, using only the digits **0** and **1**.

QUESTION 4:

Single Flip-Flop can store ____ bits of information.

- a) 2
- b) 32
- c) 64
- d) 1

Correct Answer: d

Detailed Solution:

A flip-flop is a basic building block in digital electronics, primarily used for storing a single bit of data. A single flip-flop can store only 1 bit of information, which can either be 0 or 1.

Flip-flops are bistable devices, meaning they have two stable states:

Set state (1)

Reset state (0)

Depending on the input signals and the type of flip-flop (e.g., SR, D, JK, T), it can store and maintain the bit value until it is changed by a clock signal or another input.

1 flip-flop = 1 bit of data storage.

To store multiple bits of data, a group of flip-flops (registers) is used:

For example, 8 flip-flops are needed to store 1 byte (8 bits).



QUESTION 5:

A 64 bit word consists of

- a) 4 bits
- b) 4 bytes
- c) 8 bytes
- d) 8 bits

Correct Answer: c

Detailed Solution:

8 bits = 1 byte.

$64/8 = 8$ bytes.

QUESTION 6:

What is the high speed memory between the main memory and CPU called?

- a) Magnetic Disk
- b) Cache memory
- c) DRAM
- d) Secondary memory

Correct Answer: b

Detailed Solution:

It is called the Cache Memory. The cache memory is the high speed memory between the main memory and the CPU.



QUESTION 7:

Which of the following best describes random-access memory (RAM)?

- a) A type of memory in which access time depends on memory location
- b) They lose memory when power is removed
- c) A type of memory in which access time is the same for any memory location
- d) both b & c

Correct Answer: d

Detailed Solution:

b) They lose memory when power is removed:

RAM is a type of volatile memory, meaning it requires power to retain data. When the power is turned off, the data stored in RAM is lost.

c) A type of memory in which access time is the same for any memory location:

RAM is a random-access memory, meaning any memory location can be accessed in the same amount of time, regardless of its position.

d) both b & c:

Correct: Both b and c are true descriptions of RAM.

Volatile memory: Loses data when power is removed. Random access: Any memory cell can be accessed directly and in the same amount of time. Common types include DRAM (Dynamic RAM) and SRAM (Static RAM), with DRAM being slower but more widely used in modern computers.

QUESTION 8:

CPU stands for

- a) Common purpose unit
- b) Central processing unit
- c) Central point unit
- d) common processing unit



Correct Answer: b

Detailed Solution:

QUESTION 9:

CD-ROM is known as

- a) Floppy Disk
- b) Compact Disk Read Only Memory
- c) Compressed Disk Read Only Memory
- d) Compact Disk Random Access Memory

Correct Answer: b

Detailed Solution: CD-ROM (Compact Disk Read Only Memory) is a type of optical disk that is pre-pressed with data, which means the data is written onto the disk during its manufacturing process and cannot be altered or erased.

QUESTION 10:

What is the primary function of a register in a CPU?

- a) To perform arithmetic operations
- b) To store frequently accessed data or instructions temporarily
- c) To connect the CPU to external devices
- d) To hold non-volatile data

Correct Answer: b

Detailed Solution:

*****END*****